SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	CodePulse
Project / Solution Name	MatchyKan FutureReady
Name (Leader)	Anis Sofia
Email & Phone (Leader)	anissofiamohamadsalim@gmail.com +6018-774 6081
University Affiliation(s)	University Technology Malaysia
Live Prototype URL	https://MatchyKan-CodePulse.replit.app
Code Repository URL (public / access granted to judges)	**https://github.com/anissfiaaa-beep/CodePulse_SDC_Hackathon_Submission**

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

MatchyKan FutureReady is an AI-supported career exploration and future-workforce intelligence web application. Its architecture consists of a React and TypeScript frontend, a Node.js and Express backend, an AI inference service, and a career-signal processing layer. Users can explore career matches, review skill and job-market signals, receive CV-based skill suggestions, and access an AI career coach. The frontend communicates with the backend through API endpoints. The backend retrieves public career-related news through Google News RSS feeds, processes evidence into skill signals, and maintains trend snapshots. For AI-assisted CV skill extraction and career-coach responses, the backend communicates with a RunPod Serverless endpoint running Qwen/Qwen3-8B. The application also provides career references and learning resources to support reskilling recommendations. LinkedIn and JobStreet searches help users explore related opportunities. The system is designed to support human career decisions rather than make final employment decisions. AI features require valid RunPod configuration to operate; manual skill entry remains an alternative when AI services are unavailable.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.x	Frontend and backend application logic with type checking	Apacher-2.0
React	Interactive user interface and reusable components	MIT
Vite	Frontend development server and production build	MIT

Name & Version	Purpose in Project	License
Node.js	JavaScript runtime for backend services	MIT
Express 5.x	Backend API routes and request handling	MIT
Tailwind CSS	Responsive styling and interface design	MIT
Recharts 2.15.x	Data visualization and career-signal charts	MIT
Zod	Request validation and structured data validation	MIT
Rss-parser 3.13 x	Parsing public RSS feeds for career and AI-trend evidence	MIT
Pdf-parser 2.4.x	Extracting selectable text from uploaded PDF CVs	Apache-2.0
Mammoth 1.13.x	Extracting text from DOCX CVs	BSD-2-clause
Mutler 2.5.x	Handling CV file uploads in the backend	MIT
Drizzle ORM	Database access and scheme related functionality in the workspace	Apache- 2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API	AI-powered CV skill extraction and AI career-coach responses using Qwen/Qwen3-8B	RunPod account; usage may incur charge
Google News RSS	Retrieves public news related to AI, emerging skills, cybersecurity, renewable energy, and job-market developments	Public RSS feed
LinkedIn job-search pages	Allows users to explore career opportunities related to recommended roles	Public website links; no direct recruitment API integration
JobStreet job-search pages	Allows users to explore relevant job listings and career opportunities	Public website links; no direct recruitment API integration

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development workspace and intended application deployment environment	Confirm actual account plan and deployment configuration
RunPod Serverless	Hosts and executes the Qwen/Qwen3-8B model for AI features	Serverless GPU inference; confirm endpoint region and billing plan

Service & Provider	Purpose in Project	Plan / Region
Google News RSS	Public external source for current career-related news evidence	Public feed; no paid plan declared
GitHub	Source-code repository and version control, if used for the submitted project	Confirm repository visibility and account plan

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen/Qwen3-8B	Qwen/Qwen3-8B	Pre-trained model used for CV skill extraction and AI career-coach responses	applicable RunPod service terms; verify the exact model licence before submission

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Career and AI-trend RSS evidence	Google News RSS search feeds	Public	Source publishers' applicable terms	No, when using public news articles without personal information

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Career-signal and trend snapshots	Project-generated snapshots based on collected public evidence	Self collected	Source attribution and applicable source terms	No, when restricted to non-personal career signa
FutureReady demo dataset	Built-in project demo data in the backend	synthetic	Project-created; verify attribution of any embedded third-party content	No, provided it contains only non-personal demo records
Synthetic CV test file	test-assets/synthetic-cv.txt in the project repository	Synthetic	Project-created; verify before distribution	No, provided all contents are fictional

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Public learning-resource reference	Google Machine Learning Crash Course, Kaggle Learn, Cisco Skills for All, Microsoft Learn, Open University OpenLearn, and other listed educational resources	Public references	Respective providers' terms	No, when used only as resource references

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT (GPT-6)	Assistance with code generation, debugging, technical explanations, documentation, and feature planning	Major
Replit AI tools	Website development, code assistance, and application iteration within Replit	Confirm actual usage before declaring

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team reviewed AI-assisted code and documentation, checked implementation details against the project requirements, and tested the application during development. Backend validation is used to check AI-generated skill suggestions against the application's skill catalogue. The team remains responsible for the final code, configuration, and outputs. AI-generated career guidance is intended to support exploration rather than replace human judgement. Public evidence and resource links should be checked before recommendations are relied upon. The team will correct identified errors and disclose known limitations.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud-based development environment and application hosting platform	Used to develop, test, run, and demonstrate the MatchyKan FutureReady web application. It supports the development workflow and accessibility of the prototype for judges.
RunPod	Serverless GPU infrastructure and AI model inference endpoint	Used to run the Qwen/Qwen3-8B model for AI-assisted CV skill extraction and career-coaching responses, enabling intelligent career guidance within the application.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the development and deployment environment for the prototype, while RunPod supports its AI-powered features. RunPod is particularly important for delivering AI-assisted CV analysis and career-coaching responses. Although alternative cloud hosting platforms and AI inference providers could be used, replacing these services would require configuration changes, integration work, and additional testing. The sponsor resources therefore contribute significantly to the prototype's development, accessibility, and core AI functionality

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

MatchyKan accepts career-profile information and supports PDF, DOCX, and TXT CV uploads. CV upload is optional. According to the project documentation, CV processing occurs only after user consent. The backend extracts selectable text and sends up to 20,000 characters to the configured RunPod endpoint for AI-assisted skill extraction or career guidance. The project states that uploaded CV files and extracted CV text are not persisted by the application. RunPod credentials are configured through server-side environment variables or Replit Secrets and are not exposed in frontend code. The upload size is limited to 5 MB per file. Users can review suggested skills before adding them to their profiles. The demonstration should use synthetic CVs and fictional profiles, in accordance with the Domain 1 rules. The team should verify hosting logs, third-party retention policies, and any actual profile-storage behaviour before claiming that all personal information is never stored. Users should avoid uploading real personal CVs to the hackathon demonstration.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

MatchyKan depends on the availability and quality of public RSS feeds and the configured RunPod Serverless endpoint. News coverage may be incomplete, delayed, or biased, and news frequency does not necessarily represent actual labour-market demand. AI-generated career guidance may contain inaccurate statements or suggest unsuitable skills. Future skill and role estimates are uncertain and should not be treated as guaranteed predictions. The application mitigates some risks by displaying supporting evidence and source links, distinguishing public-feed evidence from curated demo references, validating AI-extracted skills against a predefined catalogue, and providing manual skill entry when AI processing is unavailable. The backend also handles service errors and incomplete AI responses. Users should verify recommended courses and career information against the linked providers. Career matches are guidance, not hiring decisions. Further improvements should include

stronger source-quality checks, fairness testing, more Malaysia-specific labour-market data, and multiple clearly labelled future scenarios.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

24.10.2026

[Date]